

SUPERVISOR-READY MACHINE-LEARNING REPORT

CSO REGIONAL CLUSTER ANALYSIS

Predicting Collective Spill Behaviour Across WWTW Regions

A compact research report testing whether a larger, physically transparent spatial unit reveals stronger environmental signal than one selected sensor.

Core conclusion. Regional aggregation improves beta most, but it does not rescue every target. Beta reaches a 17.4% locked-test RMSE gain and R-squared 0.314; lambda and duration improve modestly; frequency is worse than Dummy. The evidence supports partial spatial-unit mismatch plus target smoothing and missing hydraulic/event-state information.

MODEL 1 Regional beta	MODEL 2 Regional lambda	MODEL 3 Pooled duration	MODEL 4 Exposure frequency
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company x 25 km target-independent WWTW clusters; shared regional catchment information

259 primary regions | 1,444 WWTWs | 2,045,832 exact assigned events 70% train | 15% validation | 15% locked test | RMSE headline metric

0. Executive interpretation: what the four regional models actually say

Target	Selected model	Test RMSE	Gain vs Dummy	R-squared	Spearman
fitted beta	ElasticNet	0.544	17.4%	0.314	0.363
inverse-timescale lambda	ElasticNet	9.822	9.7%	0.180	0.404
pooled mean duration	ElasticNet	0.979	7.8%	0.143	0.450
annualised frequency	ElasticNet	0.907	-2.0%	-0.063	0.201

The result in plain English. Changing the spatial unit helps beta most. The other positive gains are smaller and uncertain, while frequency fails on the locked test. A larger region makes targets smoother, but smoothing alone is not the same as uncovering stronger catchment physics.

Five findings to tell a supervisor first

1. Regional beta is the clearest result: 17.4% lower RMSE than Dummy, but its bootstrap interval (-8.0% to 30.1%) crosses zero.
2. Lambda and duration show modest point gains of 9.7% and 7.8%; neither is precise enough for a strong claim.
3. Frequency deteriorates by 2.0% on the locked test despite valid exposure adjustment.
4. The 50 km sensitivity often performs poorly because only 87-107 eligible regions remain.
5. The most defensible explanation combines noisy targets, partial unit mismatch, variance shrinkage, and missing hydraulic/event-state predictors.

Critical comparison rule. Compare normalized gain over each model's own Dummy, R-squared, rank correlation, target variance and uncertainty. Do not compare raw RMSE across different target definitions or spatial units.

1. How to read the metrics and plots

RMSE: the main prediction-error metric

RMSE measures the typical transformed-scale prediction error. Lower is better. Improvement is calculated relative to the internal Dummy fitted to the same target and split.

Why transformed targets? Beta and lambda are strictly positive and skewed, so log transforms prevent a few extreme regions dominating model selection. Duration and frequency use log1p.

R-squared can be negative on a locked test

Negative R-squared means a model is worse than predicting a constant mean. It is an important failure diagnostic, not an invalid number.

Spearman correlation: ranking rather than exact prediction

Spearman rho asks whether high-observed regions tend to receive high predictions. A model can rank moderately while still having poor absolute calibration.

Bootstrap intervals and calibration

Bootstrap intervals describe uncertainty in test-set improvement. Calibration slope near one is desirable; slopes well below one indicate compression toward the mean.

Most important caution. All predictor importance is conditional and predictive. It does not identify causal environmental drivers.

Residual plots

A useful model should not leave systematic curvature or funnel-shaped spread. The residual histogram reveals skew and extreme errors that an average metric can hide.

2. Experimental design: why the locked test and regional definition matter

The central methodological strength. The region definition is fixed before outcomes are inspected. Model selection happens on training CV and validation data only; the locked test is opened once after the model is frozen.

70% TRAIN	15% VALIDATION	15% LOCKED TEST
fit preprocessing and candidates	choose final specification	final evaluation once

Regional unit

No supplied authoritative EA operational- or management-catchment polygon was suitable. The documented fallback is complete-linkage clustering of WWTW centroids within company in EPSG:27700. The primary maximum separation is 25 km; 10 and 50 km are pre-specified sensitivities.

Component	Rule
Region membership	company + BNG coordinates only
Beta / lambda	sensor mean plus median/spread sensitivities
Duration	exact pooled total duration / exact event count
Frequency	exact events / monitored sensor-years
Catchment area	sum
Percentages / rainfall / slope	catchment-area weighted
Population density	population sum / represented area
Capacity	sum(load PE) / sum(design PE)

Cohort support

The primary master contains 259 regions. Median membership is 5 WWTWs and 19 assigned sensors; median event count is 3,873. Exact counts, totals and medians were reconstructed from 2,045,832 cleaned, de-duplicated assigned events.

No leakage from geography. Target values, feature values and model performance never influence the region boundary.

3. Model 1: predicting fitted beta

Question. Can regional environmental conditions predict the mean fitted beta shape parameter?

Data	225 eligible regions; 157 train, 34 validation, 34 locked test	Selected model	ElasticNet, alpha=0.01
Test RMSE	0.544	Dummy RMSE	0.659
RMSE gain	17.4%	R-squared	0.314
Rank correlation	0.363	95% gain interval	-8.0% to 30.1%

What this means

- The model reduces transformed-scale RMSE by 17.4% and has the largest regional R-squared.
- The bootstrap interval still crosses zero because only 34 regions are in the locked test.
- Population density dominates permutation importance, but this is conditional predictive evidence.

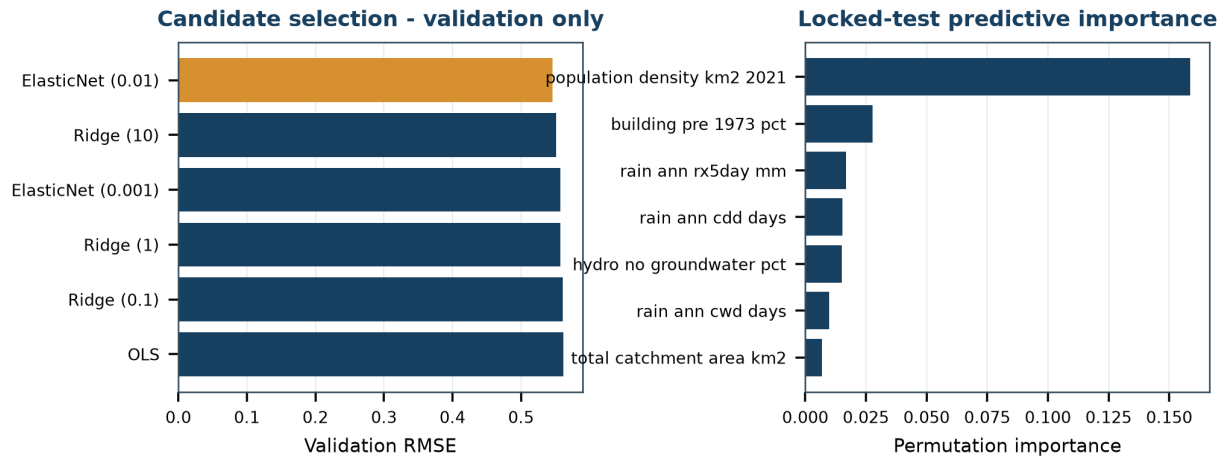


Figure. Validation-only candidate selection and locked-test predictive importance for regional fitted beta.

Core finding. Promising but uncertain regional signal.

Locked-test diagnostics: where regional fitted beta succeeds and fails

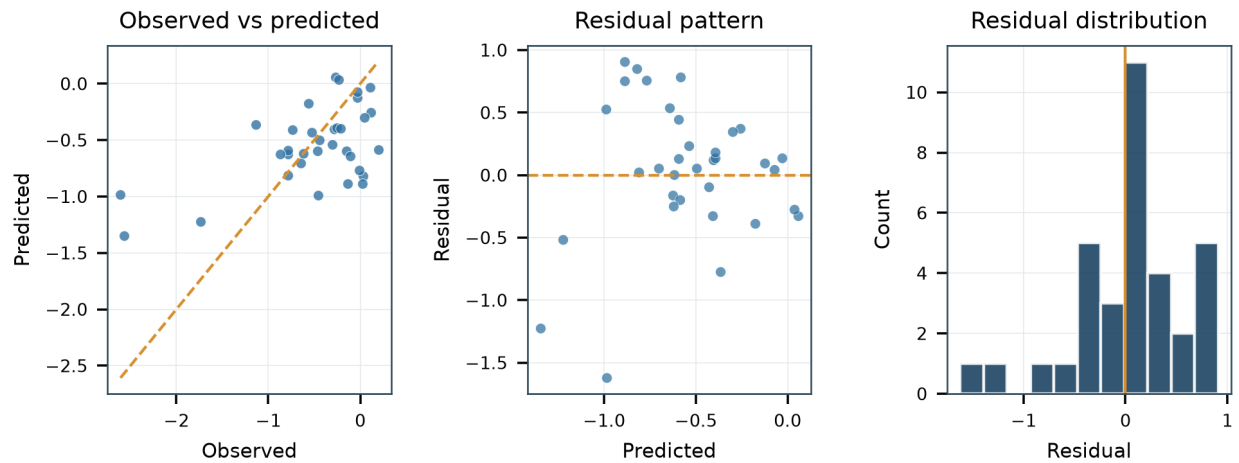


Figure. Observed versus predicted values, residual pattern, and residual distribution on the transformed scale.

How to read the diagnostics. Predictions show useful rank ordering but still compress high and low beta values. Residual spread and the wide bootstrap interval prevent a strong generalization claim.

Scientific verdict. The larger analysis unit appears to remove some selected-sensor noise, but the result is compatible with both better physical alignment and smoothing of extremes.

Guardrail. RMSE is the headline prediction-error metric. R-squared can be negative on a locked test; this correctly indicates performance worse than Dummy rather than a software failure.

4. Model 2: predicting inverse-timescale lambda

Question. Can regional conditions predict the mean inverse-timescale lambda?

Data	225 eligible regions; 157 train, 34 validation, 34 locked test	Selected model	ElasticNet, alpha=0.1
Test RMSE	9.822	Dummy RMSE	10.879
RMSE gain	9.7%	R-squared	0.180
Rank correlation	0.404	95% gain interval	-2.1% to 21.5%

What this means

- The model improves on Dummy by 9.7%, with modest rank ordering of unseen regions.
- The wide target scale and bootstrap interval show that the gain is uncertain.
- Beta and lambda are jointly fitted, so parameter instability remains relevant.

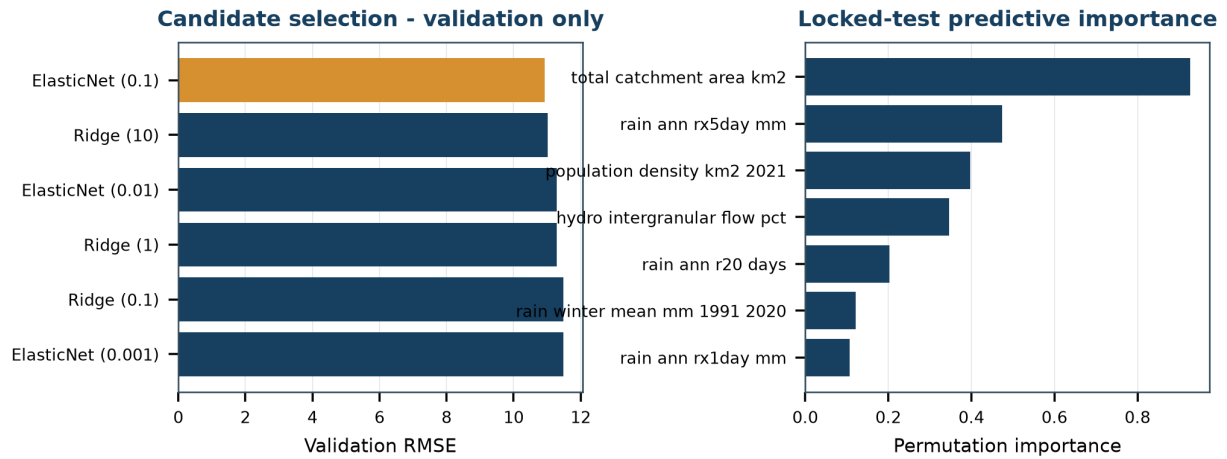


Figure. Validation-only candidate selection and locked-test predictive importance for regional inverse-timescale lambda.

Core finding. Modest, uncertain regional signal.

Locked-test diagnostics: where regional inverse-timescale lambda succeeds and fails

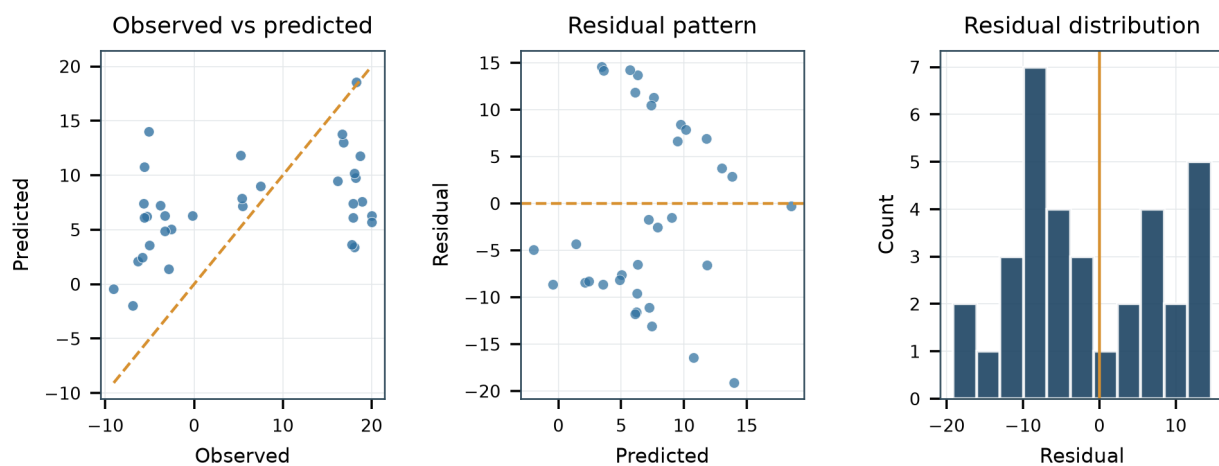


Figure. Observed versus predicted values, residual pattern, and residual distribution on the transformed scale.

How to read the diagnostics. Predictions retain modest ordering but show substantial residual spread and compression toward the mean. There is no evidence of precise regional prediction.

Scientific verdict. The model finds some aggregate signal, but richer hydraulic and event-state information is needed before the result can be treated as operationally useful.

Guardrail. RMSE is the headline prediction-error metric. R-squared can be negative on a locked test; this correctly indicates performance worse than Dummy rather than a software failure.

5. Model 3: predicting pooled mean duration

Question. Can the catchment environment predict pooled mean spill duration?

Data	257 eligible regions; 179 train, 39 validation, 39 locked test	Selected model	ElasticNet, alpha=0.1
Test RMSE	0.979	Dummy RMSE	1.061
RMSE gain	7.8%	R-squared	0.143
Rank correlation	0.450	95% gain interval	-5.4% to 17.3%

What this means

- The exact pooled mean uses total cleaned duration divided by total valid events.
- The 7.8% point gain is encouraging but its bootstrap interval crosses zero.
- Static/annual predictors remain a coarse representation of event hydraulics.

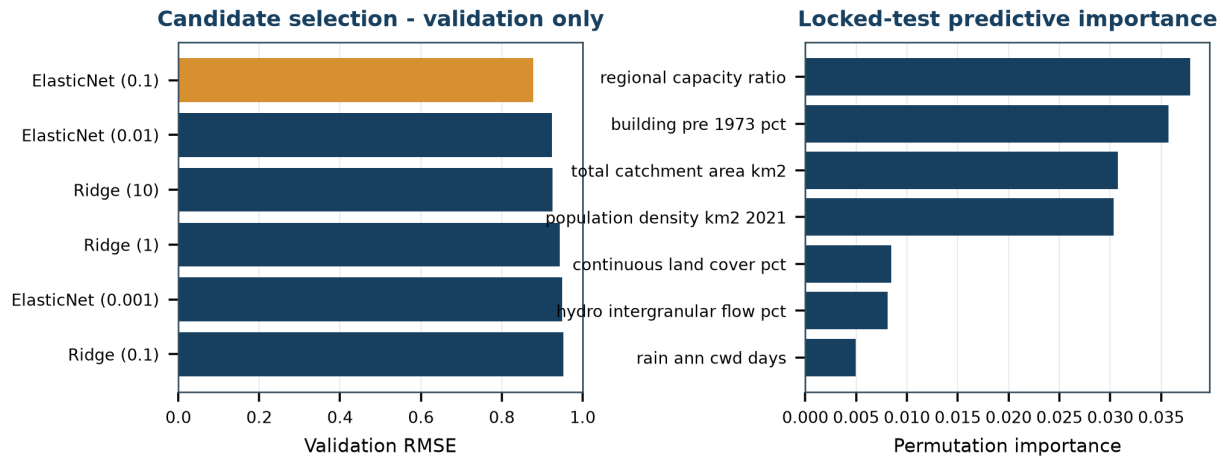


Figure. Validation-only candidate selection and locked-test predictive importance for regional pooled mean duration.

Core finding. Modest, uncertain regional signal.

Locked-test diagnostics: where regional pooled mean duration succeeds and fails

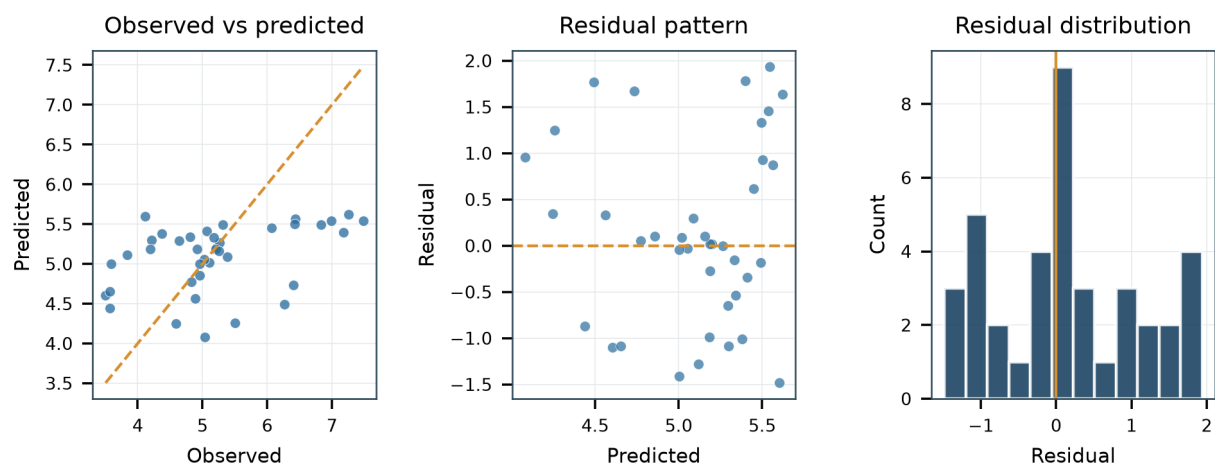


Figure. Observed versus predicted values, residual pattern, and residual distribution on the transformed scale.

How to read the diagnostics. Predictions retain modest ordering but show substantial residual spread and compression toward the mean. There is no evidence of precise regional prediction.

Scientific verdict. The model finds some aggregate signal, but richer hydraulic and event-state information is needed before the result can be treated as operationally useful.

Guardrail. RMSE is the headline prediction-error metric. R-squared can be negative on a locked test; this correctly indicates performance worse than Dummy rather than a software failure.

6. Model 4: predicting annualised frequency

Question. Can the catchment environment predict exposure-adjusted spill frequency?

Data	208 eligible regions; 145 train, 31 validation, 32 locked test	Selected model	ElasticNet, alpha=0.1
Test RMSE	0.907	Dummy RMSE	0.889
RMSE gain	-2.0%	R-squared	-0.063
Rank correlation	0.201	95% gain interval	-10.9% to 7.6%

What this means

- The locked-test model is worse than the mean-only Dummy despite good validation RMSE.
- Calibration slope is only 0.325, showing severe compression toward the regional mean.
- Exposure adjustment is correct; weak skill therefore cannot be blamed on raw count exposure.

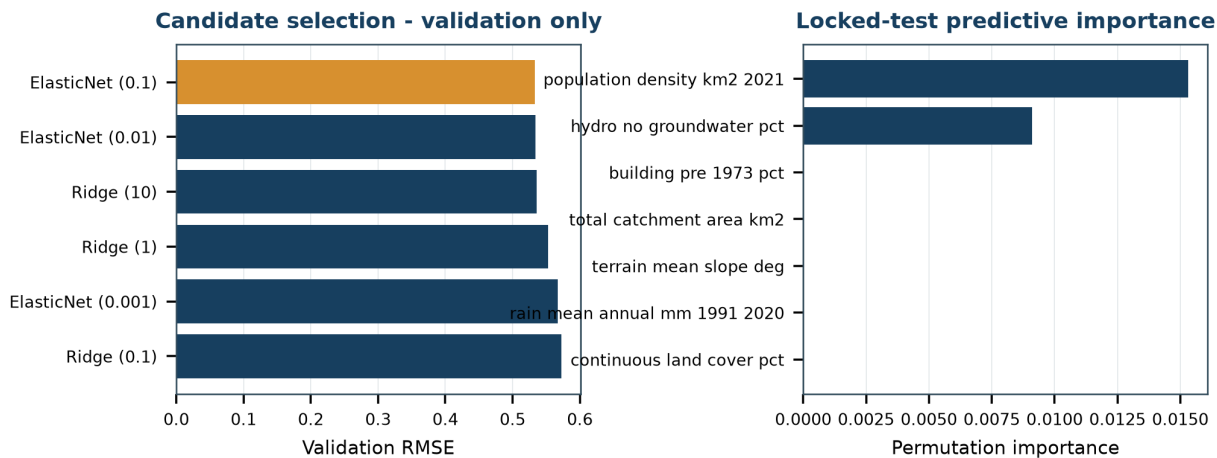


Figure. Validation-only candidate selection and locked-test predictive importance for regional annualised frequency.

Core finding. No locked-test evidence over the internal baseline.

Locked-test diagnostics: where regional annualised frequency succeeds and fails

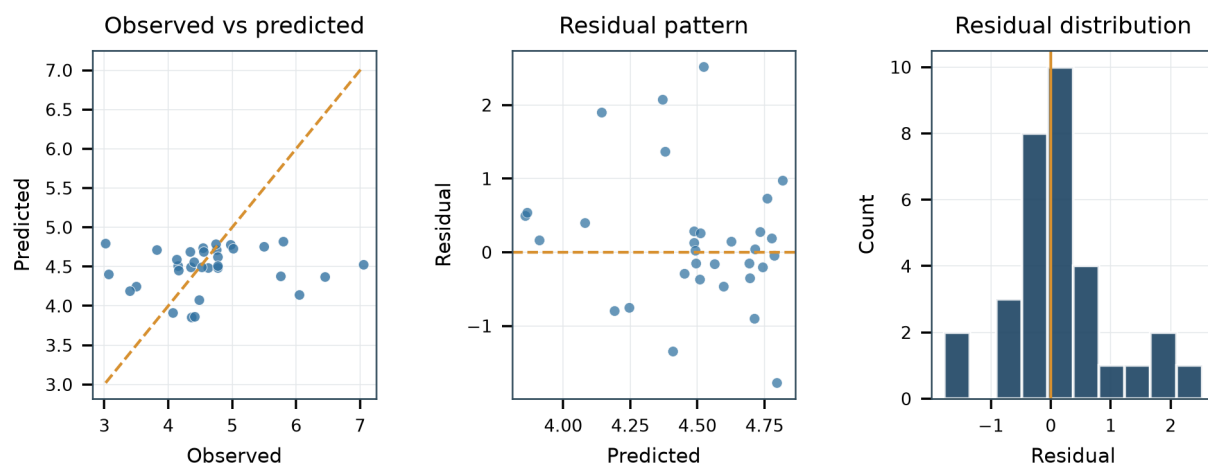


Figure. Observed versus predicted values, residual pattern, and residual distribution on the transformed scale.

How to read the diagnostics. Predictions collapse toward the centre and do not track extremes. The negative test R-squared means the selected model is less accurate than predicting the training-plus-validation mean for every locked-test region.

Scientific verdict. Regional aggregation does not rescue frequency. Annual/static predictors still omit the storm sequence, control state, storage and monitoring heterogeneity that generate spill occurrence.

Guardrail. RMSE is the headline prediction-error metric. R-squared can be negative on a locked test; this correctly indicates performance worse than Dummy rather than a software failure.

7. Spatial-scale sensitivity and old-versus-regional comparison

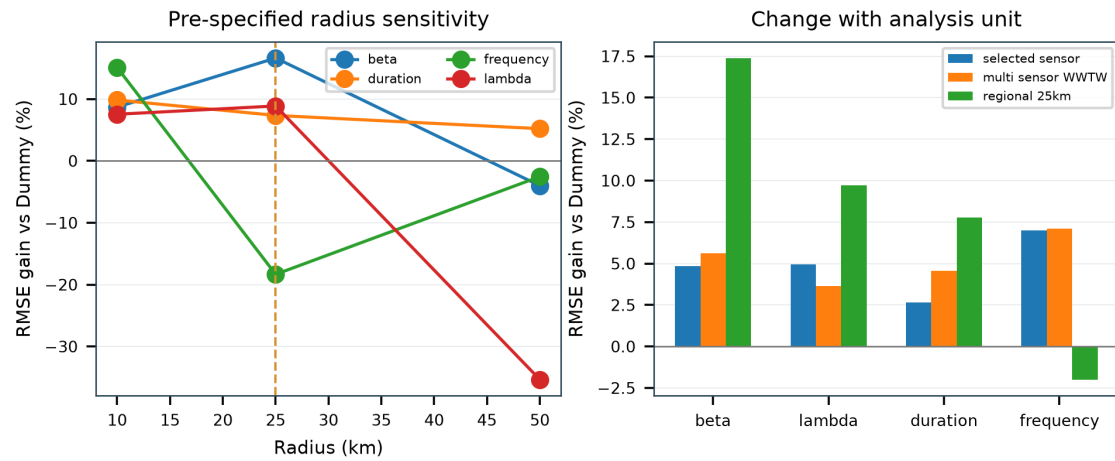


Figure. Fixed-Ridge radius sensitivity and normalized improvement for selected-sensor, multi-sensor WWTW, and primary regional units.

Radius km	Target	Regions	RMSE gain %	R-squared
10	beta	523	8.7	0.166
10	lambda	523	7.5	0.144
10	duration	638	9.8	0.187
10	frequency	506	15.1	0.276
25	beta	225	16.6	0.301
25	lambda	225	8.9	0.164
25	duration	257	7.3	0.135
25	frequency	208	-18.3	-0.431
50	beta	100	-4.0	-0.100
50	lambda	100	-35.3	-0.844
50	duration	107	5.2	0.073
50	frequency	87	-2.5	-0.383

How to interpret radius sensitivity. Ten kilometres can perform better for some targets, while 50 km often loses signal and sample size. These results do not replace the pre-declared 25 km headline; they show that spatial scale itself is a material modelling assumption.

Smoothing warning. A larger cluster can improve RMSE by shrinking extremes even when the predictor-target relationship is not physically stronger. Normalized metrics and variance change must be read together.

8. Learning curves: would simply adding more regional rows fix the problem?

Regional learning curves

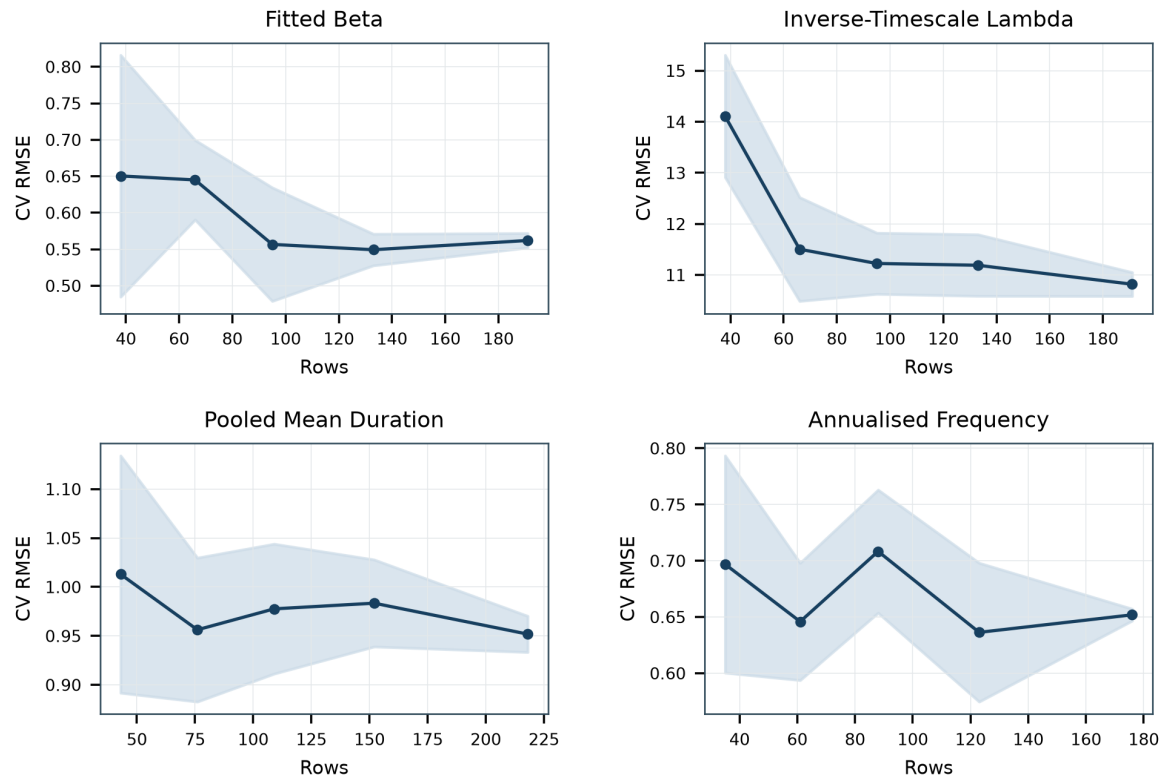


Figure. Training-plus-validation sample size against cross-validated transformed-scale RMSE.

Target	First CV RMSE	Largest-n RMSE	Change	Diagnosis
beta	0.650	0.562	13.6%	still declining
lambda	14.110	10.819	23.3%	still declining
duration	1.013	0.952	6.0%	near plateau
frequency	0.697	0.652	6.5%	near plateau

Why a plateau matters. More similar regions are unlikely to create strong prediction if feature information and target reliability are limiting. The curves argue for richer process-matched inputs before a much larger algorithm search.

9. Spatial/company robustness and the hierarchical benchmark

Spatial-block and leave-one-company-out checks

Target	100 km spatial CV RMSE	LOCO folds	Headline test RMSE
beta	0.598	9	0.544
lambda	11.298	9	9.822
duration	1.053	9	0.979
frequency	0.679	7	0.907

These sensitivities are deliberately harder than random regional prediction. Operator-specific monitoring, network practice and geography can dominate individual held-company folds.

Hierarchical benchmark

Target	Method	Rows	Regions	Status	Held-region RMSE
beta	OLS WWTW fixed effects	1008	225	converged	0.622
beta	MixedLM random intercept by region	1008	225	failed: LinAlgError: Singular matrix	0.559
duration	OLS WWTW fixed effects	1305	257	converged	1.250
duration	MixedLM random intercept by region	1305	257	failed: LinAlgError: Singular matrix	1.240

Why the mixed model is not a success claim. The full beta and duration MixedLM fits reached singular random-effect covariance. Some folds fitted, but the full specification is unstable. It is reported as an honest negative result and not promoted above regional averaging.

What would make partial pooling defensible?

Use event- or sensor-level outcomes, a better physical grouping hierarchy, explicit exposure/likelihood structure, and enough repeated observations per group. A random-intercept model should not be forced merely because partial pooling is attractive in principle.

Robustness verdict. Regional beta remains the most promising signal, but uncertainty and spatial/operator sensitivity mean it requires confirmation rather than immediate operational interpretation.

10. Why the models remain weak: parameter geometry and missing physics

Does the bounded beta scale explain part of the beta-lambda coupling?

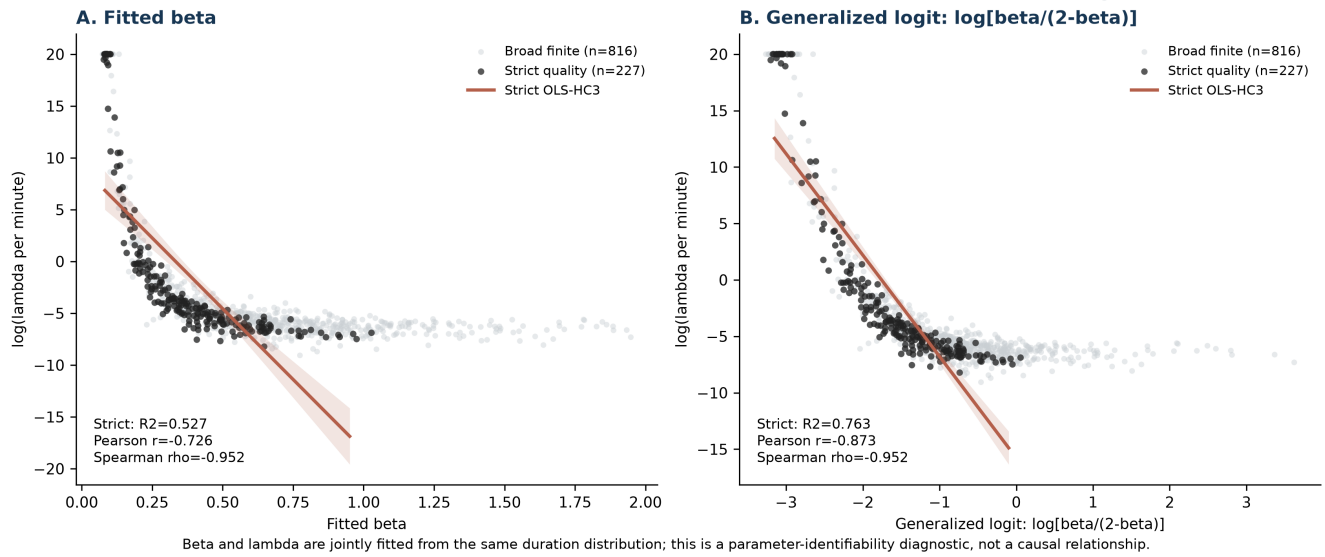


Figure. Original beta and generalized logit $\log[\text{beta}/(2-\text{beta})]$ against $\log(\text{lambda})$. Exact optimizer-boundary values are excluded rather than clipped.

Parameter-geometry finding. Strict-sample R-squared rises from 0.527 to 0.763 after the bounded-coordinate transform, but RESET and quadratic tests remain highly significant. Beta and lambda are jointly fitted from the same duration distribution, so this is an identifiability diagnostic, not a causal relationship.

Limitation	Why it matters
Target quality	tail fits and extreme estimates remain uncertain
Hydraulic state	topology, storage, pumping and control are absent
Catchment scale	environmental polygons do not equal sewer systems
Event dynamics	annual/static summaries omit storm sequence and saturation
Operator variation	monitoring and operating practice differ
Ecological aggregation	averaging can reduce variance without stronger physics

The wrong conclusion would be: The regional model has solved the problem because beta R-squared is higher. The correct conclusion is narrower: aggregation appears to remove some noise for beta, while physical information remains insufficient for strong prediction across all outcomes.

11. Supervisor-facing synthesis: how to explain the results clearly

Thirty-second summary. We replaced one selected sensor with independently defined within-company WWTW regions and retained a locked test. Regional beta improves most; lambda and duration improve modestly; frequency does not. No result is precise enough to prove a universal regional mechanism. The combined evidence points to partial unit mismatch, smoothing, fitted-parameter noise, and missing hydraulic/event-state information.

If asked: 'Did the regional unit work?'

Partially. Beta gains materially, but the confidence interval crosses zero and frequency deteriorates.

If asked: 'Did aggregation just make targets easier?'

Partly. Variance and extremes shrink, so normalized gain and locked-test diagnostics are essential.

If asked: 'Which regional model should be trusted most?'

Beta is the strongest candidate for confirmation, not deployment. Duration and lambda are qualified; frequency fails.

If asked: 'What did the hierarchical model show?'

The attempted random-intercept model was singular. Better group structure and event-level likelihoods are required.

If asked: 'What should we do next?'

Acquire topology, pipe/storage capacity, pumping/control, event rainfall, antecedent saturation, groundwater and receiving-water state before increasing algorithm complexity.

Final validation status

All 40 scientific release gates pass: exact event aggregation, unique geography, exposure handling, historical split reuse, model reloads, figure pairs and rendered-PDF checks. The workbook packaging limitation is separate from the modelling results.

Final research conclusion. Changing the spatial unit reveals a stronger beta signal but does not produce a generally strong regional prediction system. The highest-value next step is better physical and event-state data, followed by a carefully validated hierarchical or event-level model.

This report matches the compact supervisor-report visual system used by the CSO_NEW redesigned master report: dense evidence, paired diagnostics, decision boxes, and explicit plain-English interpretation.